Implementation of Subnetting in Cisco packet

Exp no: 9 Date

Aim:

Implementation of Subnetting in Cisco packet tracer simulator.

Subnetting:

→ To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch, and the router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host addresses each.

The IP addressing for the network shown is the topology can be as follows.

* Router R1:
* Gigabit Ethernet 0/0: 192.168.1.1
* Gigabit Ethernet 0/1: 192.168.2.1
* Switch S1:
* Fast Ethernet 0/1: 192.168.1.0/27
* PC1: 192.168.1.11
* PC2: 192.168.1.12
* PC3; 192.168.1.13
* Fast Ethernet 0/2: 192.168.2.0/27
* Switch S2:
* PC1: 192.168.3.11
* PC2: 192.168.3.12
* PC3: 192.168.3.13

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Configuring the devices:

- Now that we have added our devices and connected them, we can start configuring them. We will start by configuring the router. Right click on the router and select "CLI". This will open the command line interface (CLI) for the router. In the CLI, enter the following commands:

```
# enable
# configure terminal
# interface FastEthernet 0/0
# ip address {IP address} {subnet mask}
# no shutdown
# exit.

interface FastEthernet 0/1
ip address {IP address} {subnet mask}
no shutdown
exit
```

Testing the network:

Now that our network topology is configured, we can test the network. Open a command prompt on each PC and try to ping the other PC. If the ping is successful, then the network is functioning properly. We can also use the "ping" command to test connectivity between the router and the PCs.

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Fire	Last status	Source	Destination	Type	color	Time	Period-ic
●	Successful	PC4(2)	Router2	ICMP	Green	0.00(s)	N
●	Successful	PC4(2)	PC2(1)(1)	ICMP	Blue	0.00(s)	N
●	Successful	PC0	Router0	ICMP	Dark blue	0.00(s)	N

